

BACKGROUND

Social Network Analysis (SNA) is an interdisciplinary field that maps and measures relationships between people and groups. This analysis investigates the patterns of scholarly publication collaboration among professors in the college of Family, Home, and Social Sciences (FHSS) at BYU and seeks to answer the following questions:

- How do departments within FHSS differ in terms of **internal clustering**?
- How do departments differ in terms of **inter-departmental ties**? Which departments collaborate most outside of their own subgroup?

FINDINGS

Departmental Clustering

The Clustering Coefficient (0-1) measures cohesion in terms of % of *closed triads* (three fully connected individuals).

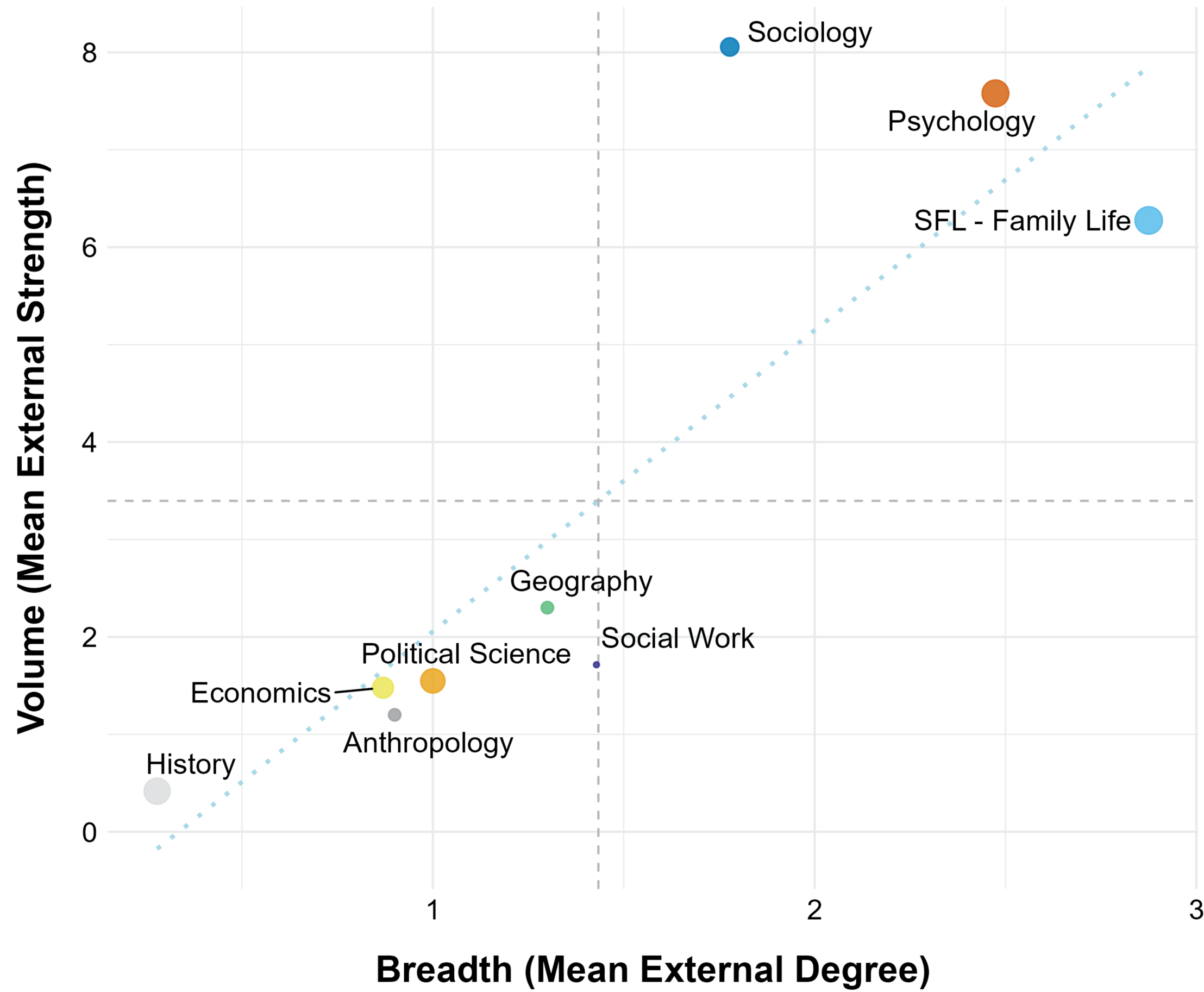
As observed in Table 1, **highest internal cohesion** is observed within Sociology, School of Family Life, and Geography. History, Anthropology, and the Dean’s Office have clustering coefficients of 0.0, indicating **low internal cohesion** in terms of closed triads.

Though clustering is not based on group size, there is a pattern of lower clustering in departments with fewer faculty. This should be noted when comparing department cohesion.

Table 1. Clustering coefficients by department.

Department	Clustering Coefficient	Size
Sociology	.56	18
SFL – Family Life	.54	40
Geography	.53	10
Social Work	.38	7
Political Science	.37	31
Economics	.33	23
Psychology	.33	38
Anthropology	.00	10
Dean’s Office	.00	3
History	.00	36
FHSS Department	.37	216

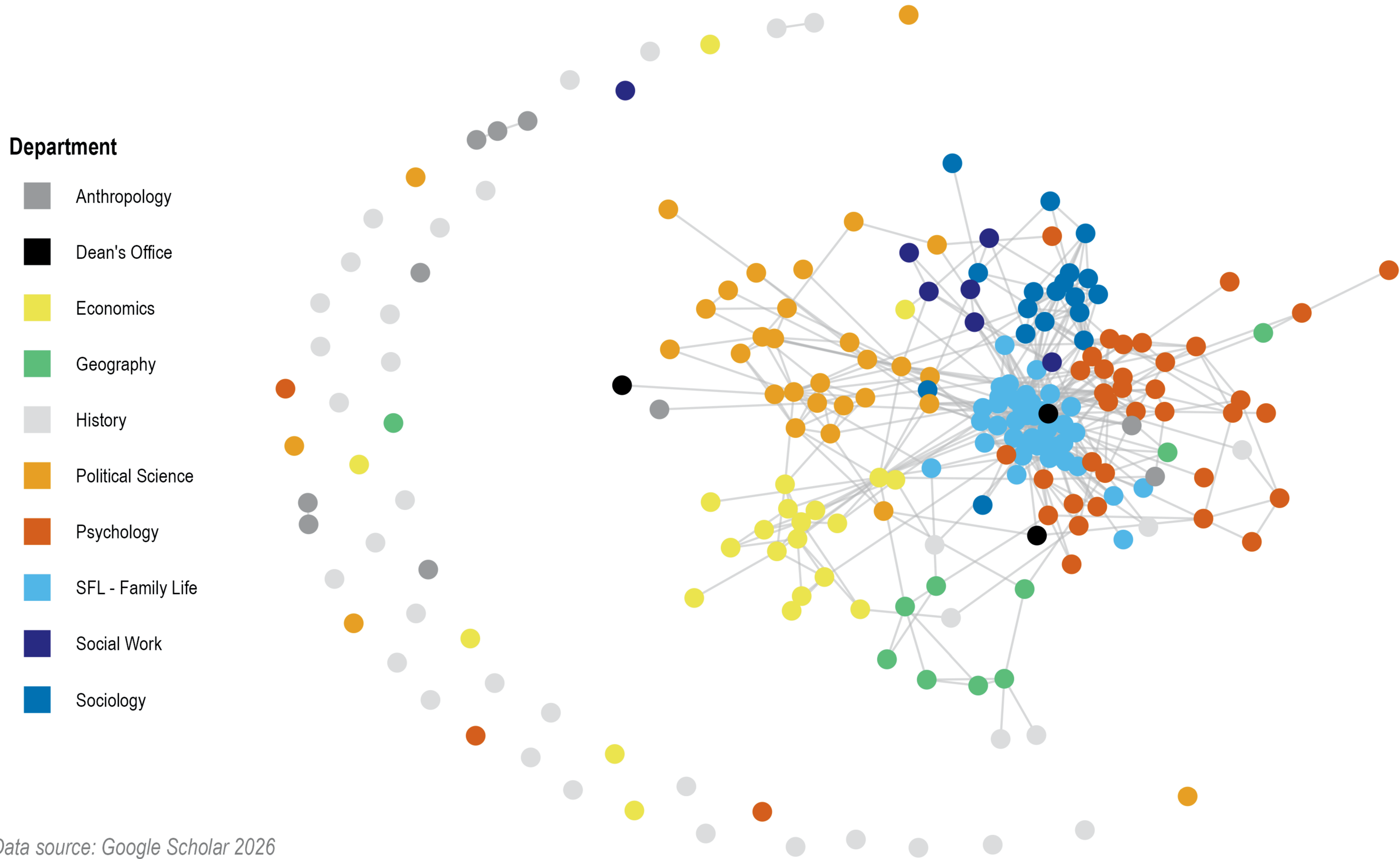
Figure 2. Average external collaboration by department.



DATA & METHODS

- A complete list of publishing BYU FHSS faculty was compiled from the FHSS website.
- A list of coauthorship ties was constructed from Google Scholar data and arranged into an edgelist, a two-column list of ties with an additional file including faculty demographics. The edgelist and vertex data were used to graph the network.
- Data analysis was conducted using R scripts. Network measures including clustering coefficient, degree, and strength were used to analyze cohesion and inter-departmental relationships.

Figure 1. FHSS co-authorship network by department.



Inter-Departmental Ties

We explored inter-departmental ties using **degree** or breadth (number of unique collaborators outside of one’s own department) and **strength** or volume (number of collaborations outside of one’s own department).

As shown in Figure 2, Sociology, Psychology, and the School of Family Life have high breadth and high strength, meaning that they collaborate most with faculty outside of their respective departments and are **bridges between the other departments** in FHSS.

The **breadth and volume of collaborative ties are observed to be related**; the more collaborations one has the more unique collaborators.